

Low-Resource Embedded DAS Beamformer for Medical Ultrasound

Federico Villani¹, Milos Hirs¹, Luca Benini^{1,2}, Andrea Cossettini¹

¹Integrated Systems Laboratory of ETH Zurich, Zurich, Switzerland, ²Department of Electrical, Electronic and Information Engineering (DEI), University of Bologna, Bologna, Italy.

Background, Motivation and Objectives

Portable US systems are increasingly integrating higher channel counts and advanced embedded acquisition capabilities. These systems require on-device image generation to enable end-users to effectively position and steer the probe during operation, after which autonomous feature extraction and downstream processing can proceed on raw data without requiring continuous image reconstruction. At the same time, embedded platforms remain constrained in computation, memory, and power, creating the need for ultra-low-resource beamforming approaches capable of low-frame-rate operation alongside the raw data acquisition pipeline, without requiring a full conventional reconstruction chain. This work presents our ongoing development toward an ultra-low-resource, I/Q-based embedded beamforming architecture for portable US acquisition systems, aiming at real-time image preview alongside acquisition and storage.

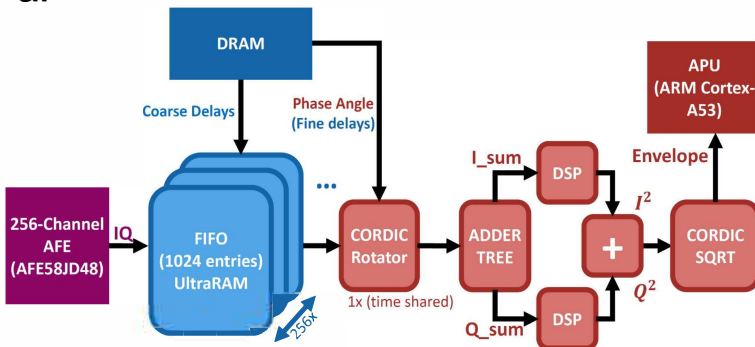
Statement of Contribution/Methods

We propose a low-resource beamformer architecture that leverages distributed computation across modern analog front-ends (AFEs) and a system-on-chip (SoC) featuring an Application Processing Unit (APU) to reduce FPGA fabric utilization. First, the AFEs acquire the data, downconvert them, and downsample them to baseband around the transducer's center frequency. The resulting I/Q samples are transmitted to the FPGA via JESD204B. On the FPGA, coarse delays programmed by the APU are applied, followed by phase shifting and apodization. The channel data are then summed, and the envelope is extracted through a CORDIC unit. The envelope data are subsequently transferred to the APU, where image reconstruction, log compression, and display are performed.

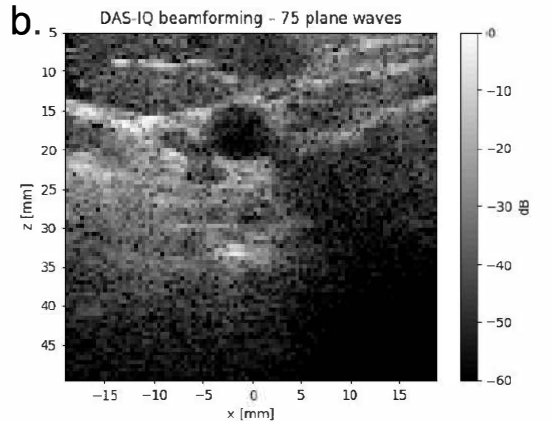
Results/Discussion

We present the first synthesis-based results of this beamformer architecture. The approach reduces FPGA utilization by shifting IQ demodulation to the AFEs and final image-processing steps to the APU. Initial synthesis results indicate operation at 300 MHz on the FPGA, supporting a 100×100 image preview at 30 FPS from 256-channel, 16-bit, 30 Msps I/Q-demodulated data. Utilization is below 10% for FFs and LUTs on a low-end ZU5EG SoC, and 2% LUTs / 1.6% FFs on a high-end ZU19EG, with BRAM usage of 8 Mb for a 256-channel implementation. Future work will focus on complete FPGA/APU integration and validation within the embedded acquisition system.

a.



b.



a. SoC reconstruction chain overview. b. DAS beamforming of PICMUS carotid scan using the low-resource beamformer.